



PROBABILITY DISTRIBUTION (V2)

MCQ SHORTCUT TRICKS — By Thunderstudy

Understand in 10 Seconds: A table of chances showing “What value can come?” + “How likely it is?”

⚡ TRICK 1 — The Golden Rule (Instant Check)

✓ **Sum of all probabilities = 1**

MCQ Shortcut: If missing 'k', subtract sum of others from 1.

$$k = 1 - (\text{sum of remaining probabilities})$$

Example: $0.2 + 0.3 + k = 1$ ➡ $k = 0.5$. **Done in 5 seconds.**

⚡ TRICK 2 — Expected Value (Mean) SUPER FAST

Shortcut Result: Multiply row-wise → Add directly.

X	P(X)	XP(X)
1	0.2	0.2
2	0.5	1.0
3	0.3	0.9

Mean = $0.2 + 1.0 + 0.9 = 2.1$

TRICK 3 — Valid Distribution Check

A table is valid ONLY IF: 1. Probabilities are 0 to 1, and 2. Total Sum = 1.

3-Second Elimination: Negative value? >1? Sum $\neq$ 1?  **Eliminate instantly!**

⚡ TRICK 4 — Variance Without Panic 😎

Formula:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Steps: 1. Add $X^2P(X)$ column to get $E(X^2)$ | 2. Subtract Mean^2 . **No long formulas.**

⚡ TRICK 5 & 6 — SD & Option Elimination Strategy

SD Hack: Find variance roughly and take the root mentally to pick the nearest option.

Topper Trick: Mean must lie between the minimum and maximum X . If X is 1-4, Mean cannot be 0 or 5!

⚡ TRICK 7 — Probability Pattern Shortcut

If probabilities increase with X (e.g., 0.1, 0.2, 0.3, 0.4), the **Mean will be towards the larger values**. Use this for estimation MCQs!

🧩 5-Step MCQ Attack Strategy

1. Check sum = 1.
2. Eliminate wrong options using Range & Sum tests.
3. Multiply $X \times P(X)$ quickly row-by-row.
4. Add column vertically for the Mean.
5. Estimate before selecting the final answer.

🕒 **Average time: 20–30 seconds per MCQ**



ULTRA-SHORT MEMORY FORMULA

“Multiply → Add → Square → Subtract → Root”

$XP(X) \rightarrow \text{Mean} \rightarrow \text{Mean}^2 \rightarrow \text{Variance} \rightarrow \text{SD}$

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